

表 7: 不同模型在多个数据集上的异常检测结果

模型	FlowGraph							TraceLog						
	AUROC	AP	P@1%	R@1%	TPR@1%FPR	F1	MCC	AUROC	AP	P@1%	R@1%	TPR@1%FPR	F1	MCC
DeepTraLog	0.9785 ± 0.0228	0.9856 ± 0.0135	1.0000 ± 0.0000	0.0300 ± 0.0000	0.9667 ± 0.0289	0.9796 ± 0.0136	0.9644 ± 0.0235	0.6472 ± 0.0163	0.5416 ± 0.0430	0.6667 ± 0.5774	0.0193 ± 0.0167	0.0694 ± 0.0673	0.1228 ± 0.1171	0.1210 ± 0.1666
GLocalKD	0.9561 ± 0.0082	0.9738 ± 0.0025	1.0000 ± 0.0000	0.0300 ± 0.0000	0.9500 ± 0.0000	0.9727 ± 0.0029	0.9526 ± 0.0054	0.6795 ± 0.0060	0.6318 ± 0.0057	1.0000 ± 0.0000	0.0289 ± 0.0000	0.2912 ± 0.0089	0.4388 ± 0.0152	0.4089 ± 0.0278
HGT	0.5374 ± 0.4560	0.4960 ± 0.1967	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	0.0000 ± 0.0000	-0.0693 ± 0.0745	0.6583 ± 0.0582	0.5532 ± 0.0662	0.5833 ± 0.3643	0.0169 ± 0.0105	0.1340 ± 0.1127	0.2277 ± 0.1195	0.2557 ± 0.1014
HRA-GNN	0.6646 ± 0.5489	0.7462 ± 0.4131	0.6667 ± 0.5774	0.0200 ± 0.0173	0.6333 ± 0.5485	0.6463 ± 0.5597	0.6110 ± 0.5810	0.8150 ± 0.0123	0.7497 ± 0.0225	1.0000 ± 0.0000	0.0289 ± 0.0000	0.2969 ± 0.0675	0.4482 ± 0.0790	0.4341 ± 0.0663
HRGCN	0.9333 ± 0.1155	0.8672 ± 0.2300	0.6667 ± 0.5774	0.0200 ± 0.0173	0.6667 ± 0.5774	0.6650 ± 0.5759	0.6438 ± 0.6093	0.7381 ± 0.0441	0.6961 ± 0.0321	1.0000 ± 0.0000	0.0289 ± 0.0000	0.3365 ± 0.0051	0.4795 ± 0.0318	0.4658 ± 0.0277
OCHetGCN	0.9557 ± 0.0561	0.8898 ± 0.1494	0.6667 ± 0.5774	0.0200 ± 0.0173	0.4200 ± 0.5188	0.5299 ± 0.5007	0.4754 ± 0.5257	0.7062 ± 0.0603	0.6015 ± 0.0551	0.6917 ± 0.2898	0.0200 ± 0.0084	0.1015 ± 0.1302	0.2167 ± 0.1662	0.2296 ± 0.1674